NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-22885/109A(USAF)
16 FEB 1990

SUPERSEDING
MIL-S-22885/109(USAF)
23 November 1987

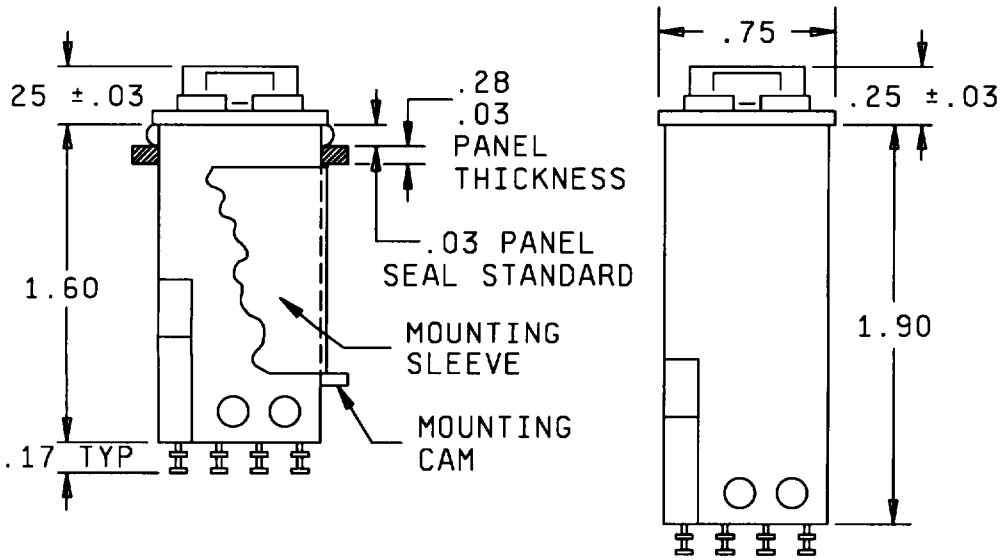
PERFORMANCE SPECIFICATION SHEET

Ⓐ

SWITCHES, PUSH BUTTON, ILLUMINATED, 4-LAMP, .75 SQUARE, 4PDT AND DPDT, 7 AMPERES SUNLIGHT READABLE DISPLAY OPTION, NVIS COMPATIBLE OPTION, MOMENTARY AND ALTERNATE HOLDOWN, LOW TOUCH TEMPERATURE, CONNECTOR MODULE TERMINATION OPTION

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-22885.



Inches	mm
.03	0.8
.17	4.3
.25	6.4
.28	7.1
.75	19.
1.60	40.6
1.90	48.3

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Push button height dimension is measured at center of legend area.
4. See figure 10 for additional applicable notes.

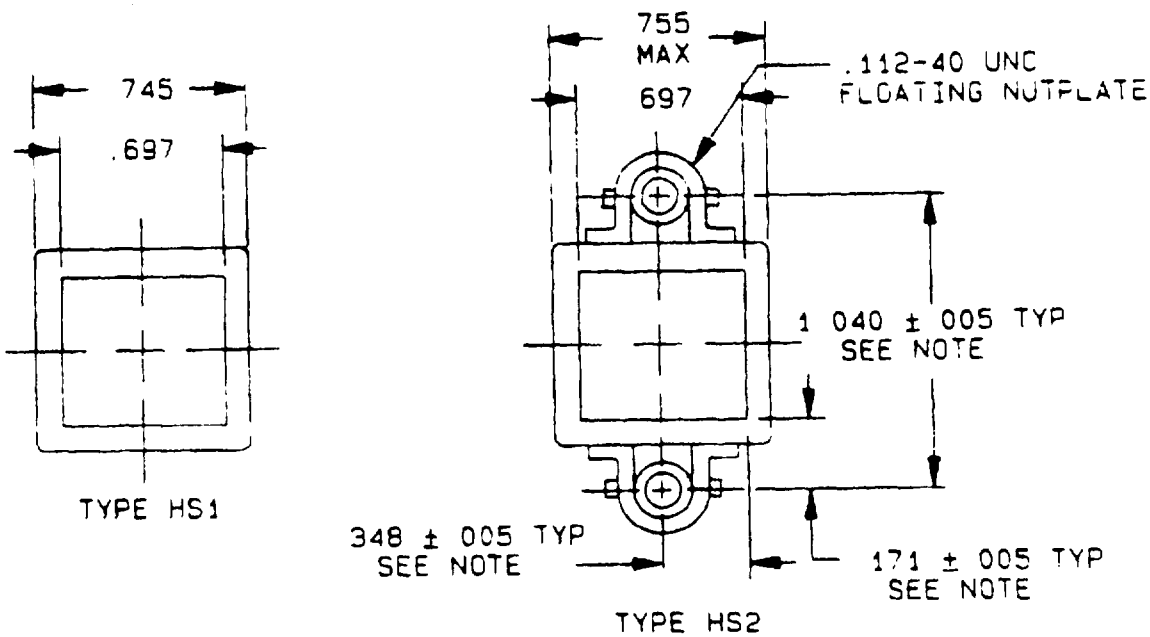
Ⓐ FIGURE 1. Four-pole switch - solder terminals.

Ⓐ denotes changes

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TOP VIEW (MOUNTING SLEEVES ONLY)

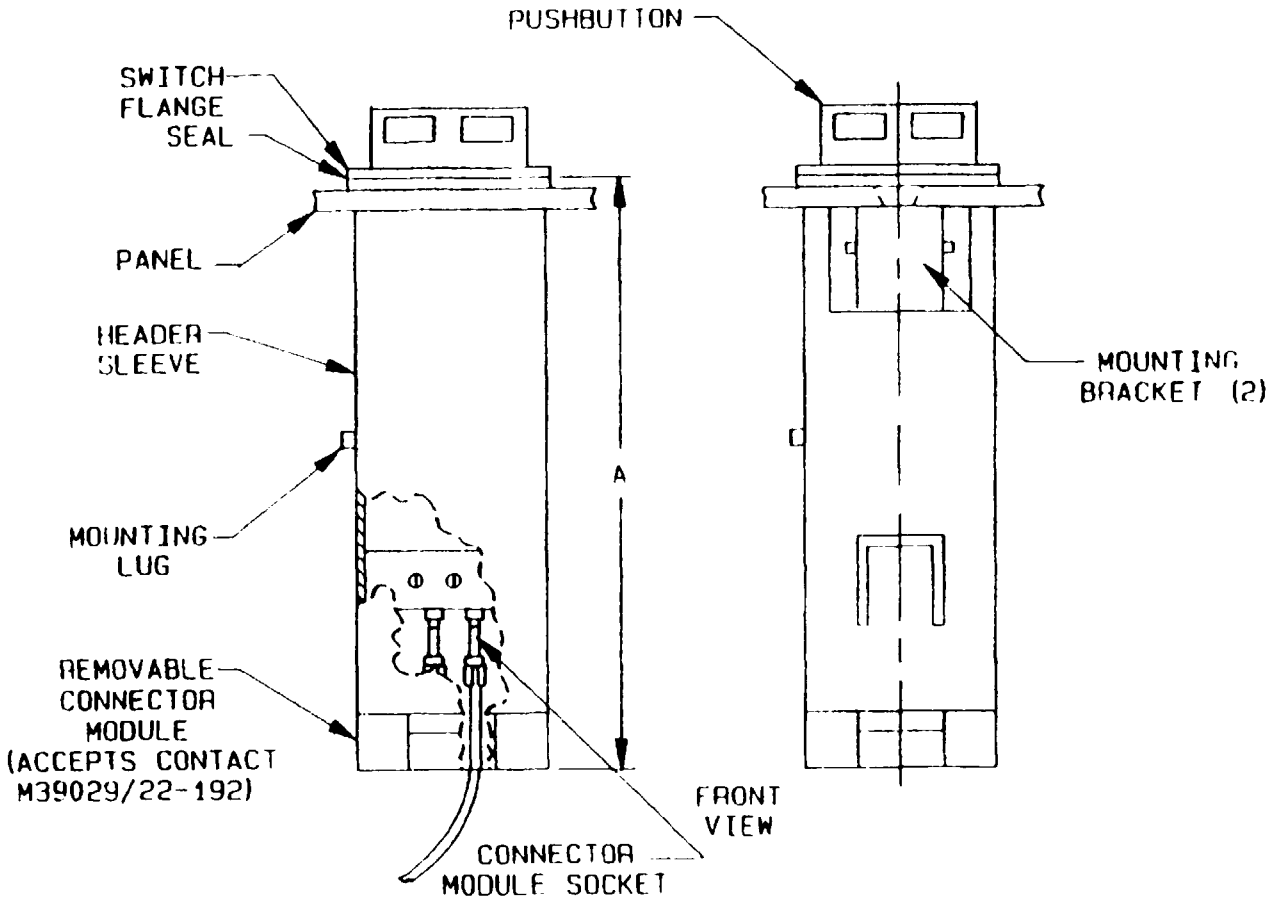
Inches	mm
.005	0.13
.112	2 84
171	4.34
.348	8.84
.697	17.70
.745	18.92
.755	19.18
1.040	26.42
1.90	48.3
2.20	55.9
2 50	63.5
2.80	71.1

Base type	Dimension A inches max
Type 1 - 4 poles (accepts short push button)	2.50
Type 2 - 4 poles (accepts high efficiency push button)	2 80
Type 3 - 2 poles (accepts short push button)	1 90
Type 4 - 2 poles (accepts high efficiency push button)	2.20

NOTE Mounting nuts float to include shown tolerances

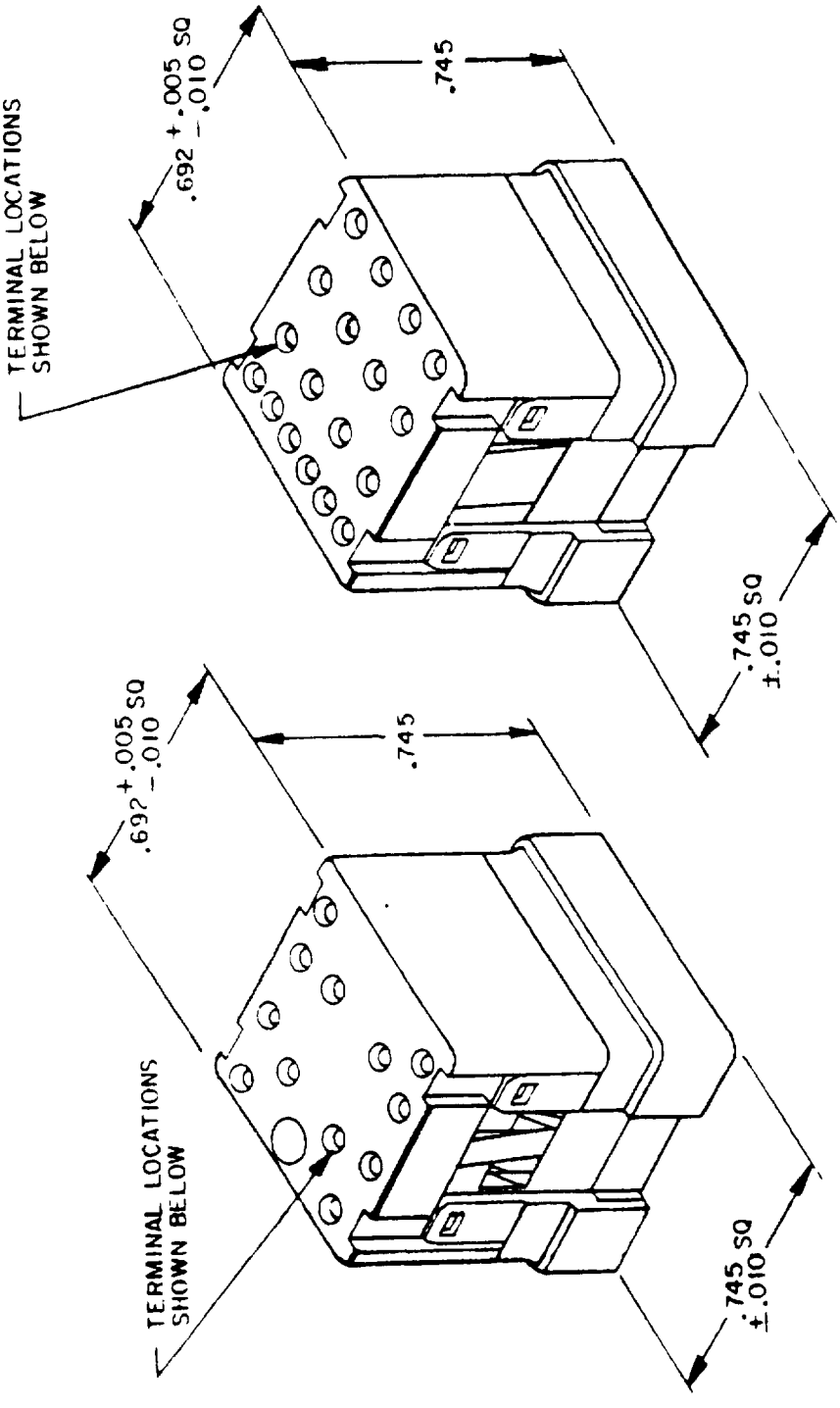
(A) FIGURE 3 Switches with connector module termination.

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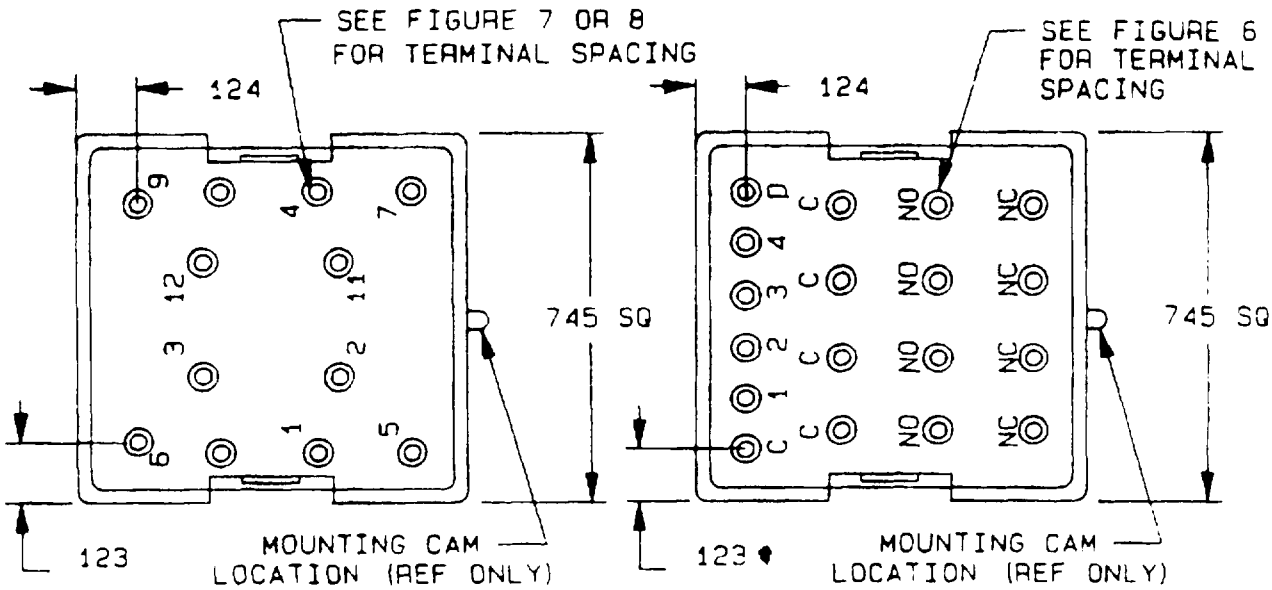
(A) FIGURE 3. Switches with connector module termination - Continued.

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(A) FIGURE 3 Switches with connector module termination - Continued

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TWO-POLE/INDICATOR
TERMINAL LOCATIONS

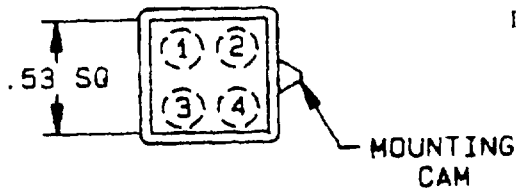
FOUR-POLE
TERMINAL LOCATIONS

Inches	mm
005	0.13
.010	0.25
.123	3.12
.124	3.15
.692	17.58
.745	18.92

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Switches with HS2 mounting sleeve are totally front panel replaceable.
4. Switches with HS2 mounting sleeve shall be provided with mounting screws as specified in table XIII.
5. The connector module shall be capable of receiving M39029/22-192 contacts crimped to a 20, 22, or 24 gauge wire.
6. The connector module shall be capable of having the socket contacts inserted or removed using an M83723/31-20 contact insertion/removal tool.
7. See figure 10 for additional applicable notes.

(A) FIGURE 3. Switches with connector module termination - Continued.

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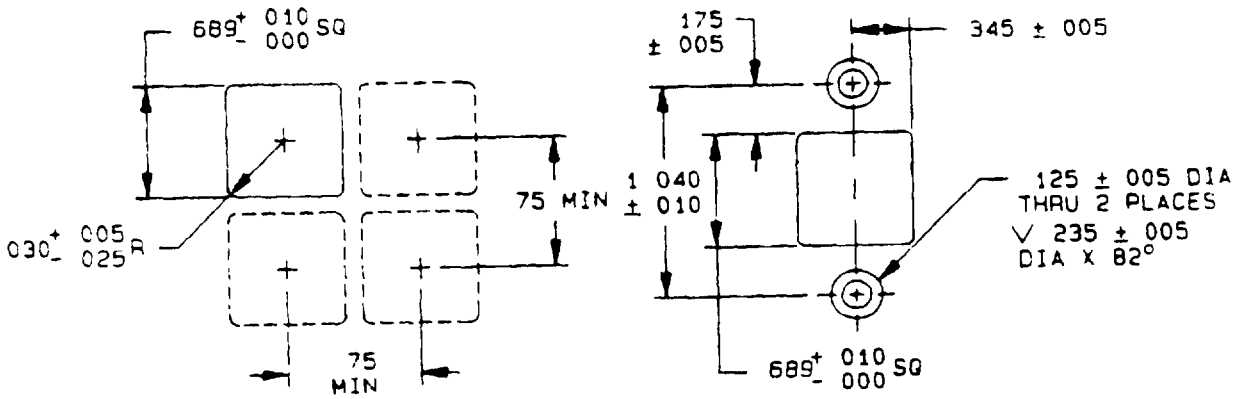


Inches	mm
53	13 5

LEGEND AREA
(FULL DISPLAY)

NOTES.
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. See figure 10 for additional applicable notes.

(A) FIGURE 4. Legend display area.

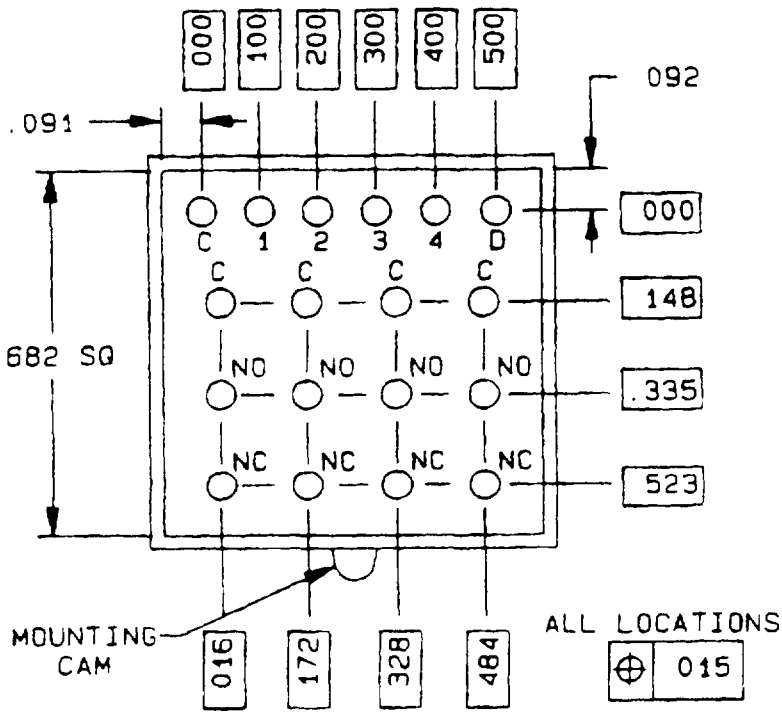


NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. See figure 10 for additional applicable notes.

Inches	mm
.005	0 13
.010	0.25
.025	0 64
.030	0.76
.125	3.18
.175	4.44

FIGURE 5. Recommended panel cutouts.

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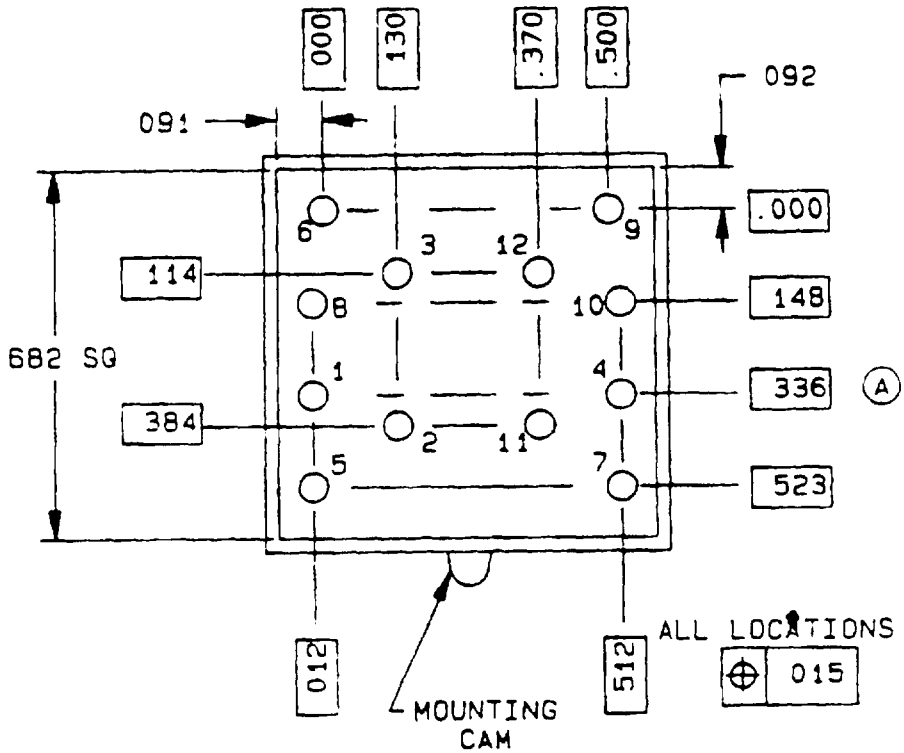


Inches	mm	Inches	mm
.015	0 38	.300	7.62
.016	0 41	.328	8.33
.091	2.31	.335	8.51
.092	2.34	.400	10.16
.100	2.54	484	12.29
.148	3.76	.500	12.70
.172	4.37	.523	13.28
.200	5.08	682	17 32

NOTES:
1. Dimensions are in inches
2. Metric equivalents are given for general information only
3. Dimensions apply for PC and connector module terminals only.
4. See figure 10 for additional applicable notes.

(A) FIGURE 6. Four-pole terminal locations

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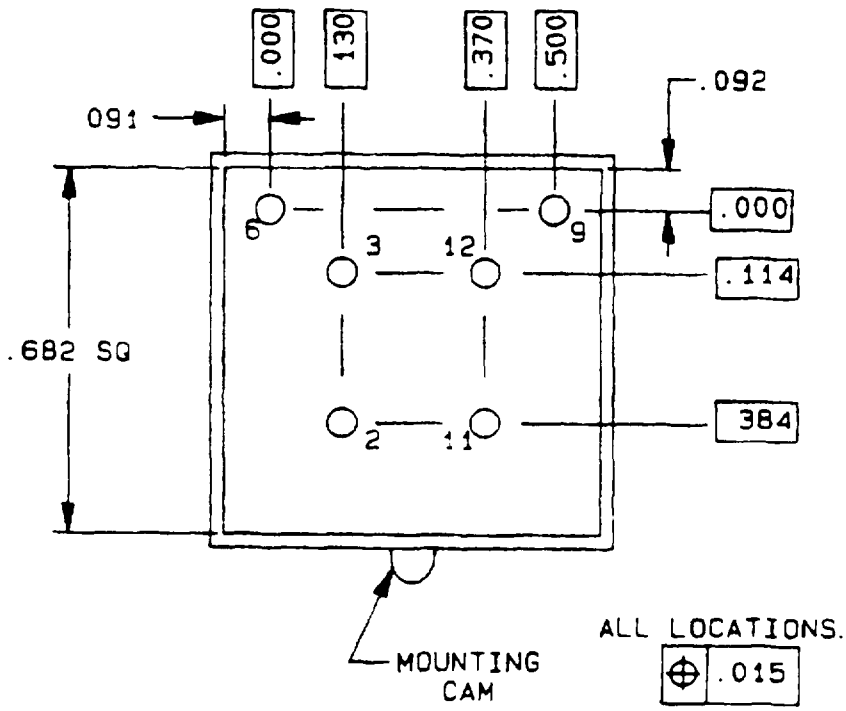


Inches	mm	Inches	mm
.012	0.30	.336	8.53
015	0.38	.370	9.40
.091	2.31	.384	9.75
.092	2.34	.500	12.70
.114	2.90	.512	13.
130	3.30	.523	13.28
.148	3 76	.682	17.32

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensions apply for PC and connector module terminals only.
4. See figure 10 for additional applicable notes.

(A) FIGURE 7 Two-pole terminal locations.

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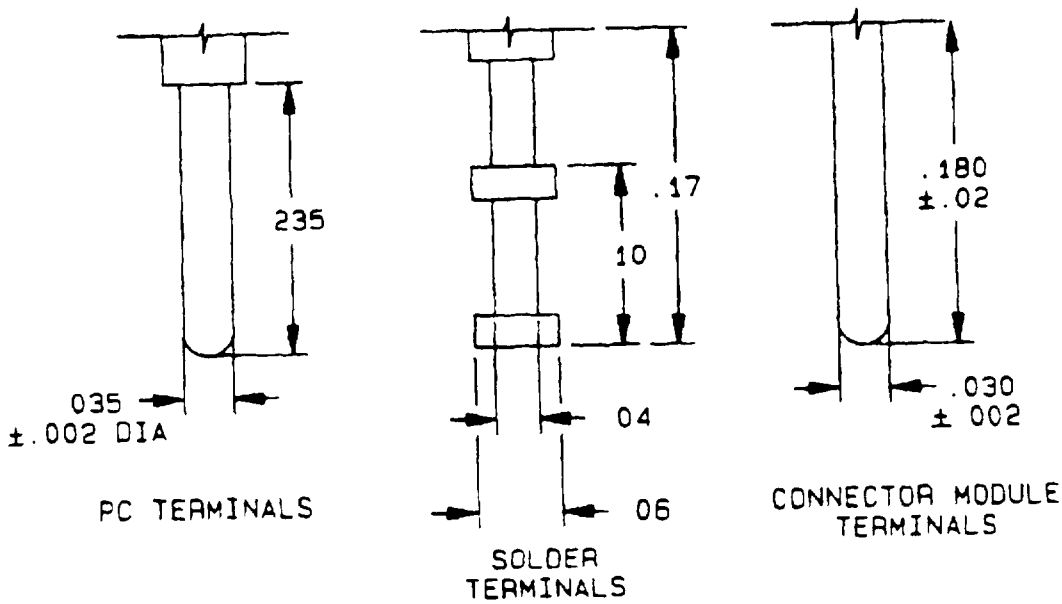


Inches	mm
.015	0.38
.091	2.31
.092	2.34
.114	2.90
.130	3.30
.370	9.40
.384	9.75
.500	12.70
.682	17.32

NOTES.
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensions apply for PC and connector module terminals only.
4. See figure 10 for additional applicable notes.

(A) FIGURE 8. Indicator terminal locations.

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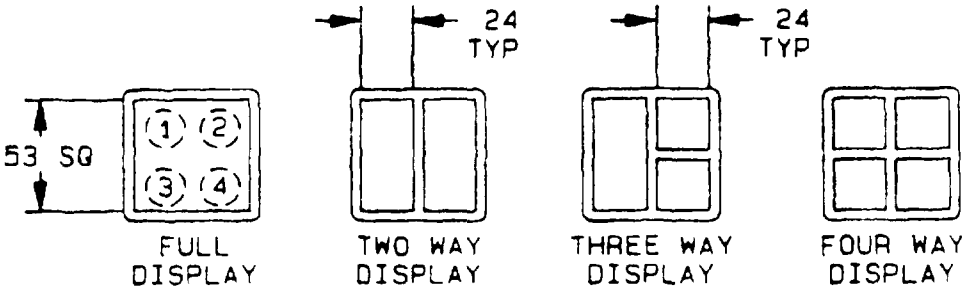


Inches	mm
.002	0.05
.02	0.5
.030	0.76
.035	0.89
.04	1.
.06	1.5
.10	2.5
.17	4.3
.180	4.57
.235	5.97

NOTES.
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. See figure 10 for additional applicable notes.

(A) FIGURE 9. Terminals.

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Inches	mm
.24	6.1
.53	13.5

NOTES.
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.015 (0.38 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals
4. The push button is designed to prevent incorrect insertion into the switch housing.
5. The push button is held captive to the switch body by a retaining element to prevent accidental interchange.
6. The exact shape of the switch is optional provided the dimensions specified are not exceeded.

(A) FIGURE 10. Lens configuration

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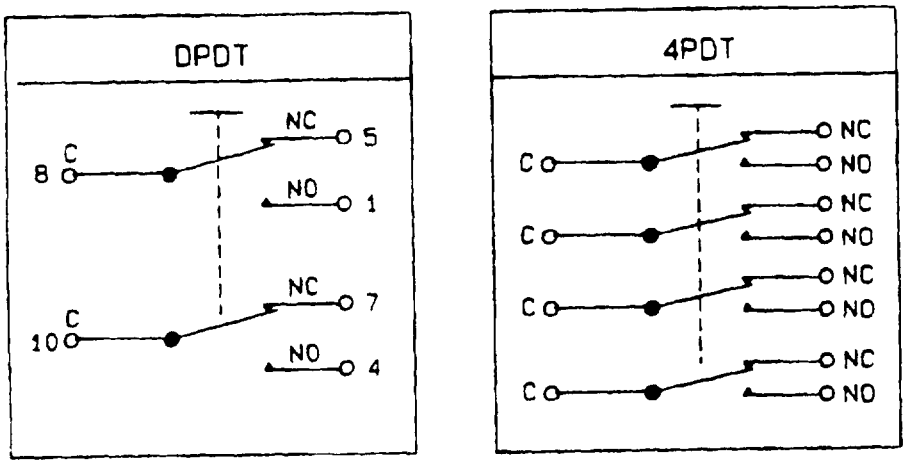
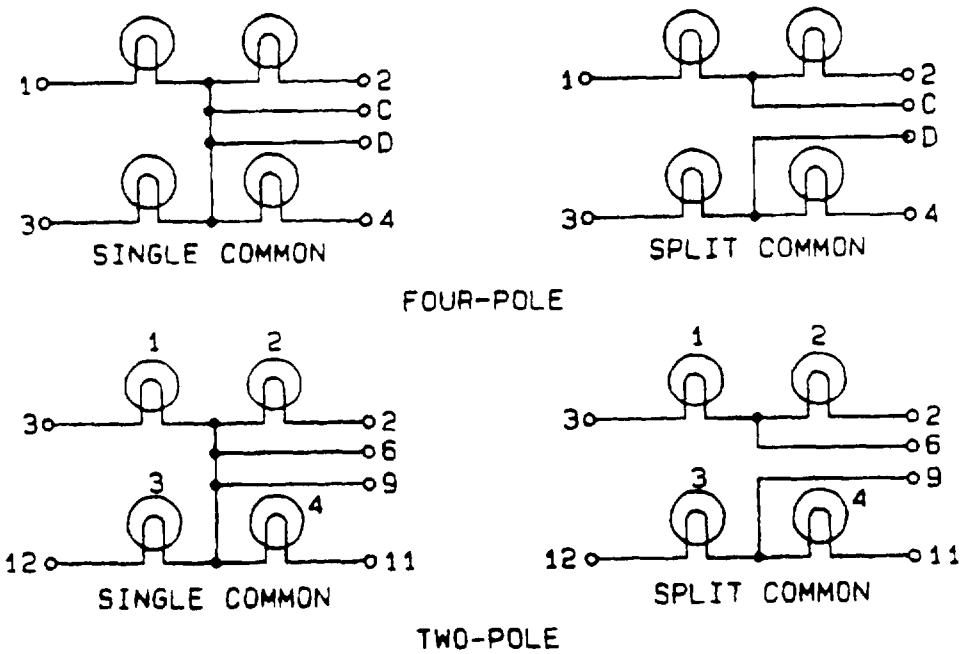


FIGURE 11. Switch arrangements.



Ⓐ FIGURE 12. Lamp circuits.

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REQUIREMENTS.

Ⓐ Design and construction. See figures 1 through 12. Basic switch shall be in accordance with MIL-S-8805/4 (MS24547-4 or -7) or MIL-S-8805/109 (M8805/109-2 or -4) listed on QPL-8805. The basic switch elements may be selected items, with reduced operating point tolerance, or category II items with printed-circuit terminals or connector module.

Enclosure design. 2 (dripproof).

Temperature characteristic. 1 (-55°C to +85°C).

Vibration grade: 3 (10 to 2,000 Hz, 15 g peak).

Operation A (momentary action), E (position indicating, alternate action), H (indicator light only).

Ⓐ Illuminated color - standard displays, incandescent illumination: See table I for standard display colors using incandescent lamps.

Ⓐ TABLE I. Illuminated color - standard displays, incandescent illumination.

Color	X 1/	Y 1/
Red (R)	.695	.285
	.703	SL 2/
	.655	.325
	.660	SL 2/
Green (G)	.300	SL 2/
	.300	.600
	.380	.600
	.380	SL 2/
Yellow (Y)	.562	.415
	.570	SL 2/
	.596	.382
	.605	SL 2/
Blue (B)	.230	.420
	.230	.350
	.320	.350
	.320	.420
White (W)	.400	.420
	.460	.420
	.400	.380
	.460	.380

1/ Chromaticity is expressed as "X" and "Y" on the CIE chromaticity diagram. Values shown are corners of limiting envelope.

2/ SL-Spectrum locus where intersected by a line of x-value shown.

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Display types: See table II.

TABLE II. Display types.

Description	Unlighted	Lighted	Type designation
Hidden legend sunlight readable	Black background	Colored lighted letters, black background	S
Hidden legend lighted background	Black background	Black letters, lighted colored background	B
Lighted background	Black letters, white background	Black letters, lighted colored background	W
Colored background	Black letters, colored background	Black letters, colored lighted background	C
Lighted letters	White letters, black background	Colored lighted letters, black background	N

Ⓐ Weight.

Four-pole with standard PB and 1.60-inch base, 0.875 ounces maximum.
Four-pole with high efficiency PB and 1.90-inch base, 1.05 ounces maximum.
Two-pole with standard PB and 1.00-inch base, 0.525 ounces maximum.
Two-pole with high efficiency PB and 1.30-inch base, 0.7 ounces maximum.
Connector module and hard-mount sleeve, 1.4 ounces maximum.

Ⓐ Lamps. Illumination is by four T-1 submidget flanged base, .15 MSCP, incandescent lamps. See application information regarding use of these switches with T-1 LEDs.

Operating characteristics.

Actuation force. 1 to 5 pounds
Actuation travel. .250 inch maximum.
Extraction force: 5 pounds maximum.

Shock: Method I (75 g, half-sine).

Electrical endurance: See table III.

Ⓐ Luminance - standard displays with incandescent illumination: See table IV for standard displays with incandescent illumination.

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TABLE III. Electrical ratings.

	Voltage	Contact rating	
		Sea level	
Silver contacts	115 V ac Resistive Inductive Lamp	7 amperes 4 amperes 2 amperes	
	28 V dc Resistive Inductive Lamp	Sea level 7 amperes 4 amperes 2.5 amperes	50,000 feet 4 amperes 2.5 amperes 2.5 amperes
Gold contacts	28 V dc Resistive Inductive	1 ampere .5 ampere	1 ampere .5 ampere

Ⓐ TABLE IV Luminance - standard displays with incandescent illumination.

Color	Minimum average brightness	
	Legend type code (see table II)	Foot lamberts
Yellow	N	300
Red		120
Green		110
White		175
Blue		90
Yellow	W	300
Red		120
Green		110
White		175
Blue		90
Yellow	S	350
Red		185
Green		185
White		185
Blue		150
Yellow	C	300
Red		120
Green		110
White		175
Blue		90
Yellow	B	300
Red		150
Green		150
White		150
Blue		100

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(A) EMI/RFI shielding. The shielding efficiency test is applicable. The shielding attenuation shall be not less than specified on figure 13.

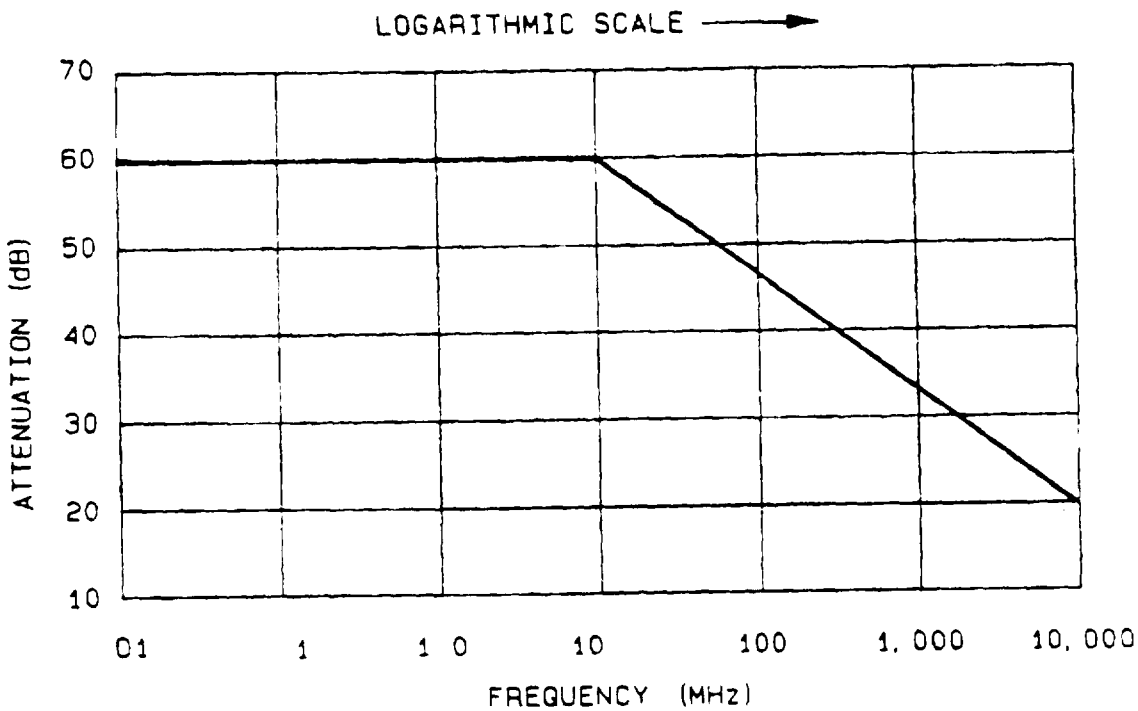


FIGURE 13. RFI/EMI attentuation.

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Ⓐ Sunlight readability: Applicable when specified only for switches with high efficiency (type 2) push button and incandescent illumination. See table V for minimum average contrast ratios in the lighted condition. In the unlighted condition, the average contrast ratio of each legend character to the background shall have an absolute value less than or equal to 0.1 when measured in accordance with MIL-S-22885.

Ⓐ TABLE V. Contrast ratios for high efficiency push button with sunlight readable display-incandescent illumination. 1/

Color	$C_L = (B_2-B_1)/B_1$ 2/ 3/	
	Condition 1 4/ ($\phi_1 = \phi_2 = 15$ degrees)	Condition 2 4/ ($\phi_1 = \phi_2 = 30$ degrees)
Red	0.6 min	0.3 min
Green	0.6 min	0.3 min
Yellow	0.6 min	0.4 min
Blue	0.4 min	0.2 min
White	0.6 min	0.2 min
NVIS green A	0.6 min	N/A
NVIS green B	0.6 min	N/A
NVIS yellow	0.6 min	N/A

1/ Application notes.

In addition to luminance contrast, color contrast is an important factor in readability. Colors in table are listed in order of general preference. Red has the best color contrast during sunlight conditions, blue and white have the poorest color contrast. Colors are in accordance with table I.

Minimum contrast values for a nonglare sunlight condition ($\phi_2 = 45$, $\phi_1 = 0$) are.

Red	0.8
Green	0.8
Yellow	0.8
Blue.	0.6
White:	0.8

Lamps with a mean spherical candlepower of .15 must be used to obtain the contrast values specified above. The qualified source(s) should be contacted for lamp recommendations for the specific equipment application.

2/ C_L = Lighted contrast.
B_2 = Legend character brightness.
B_1 = Legend background brightness.

3/ Minimum average contrast of each legend character using incandescent lamps.

4/ The photometer and high intensity light source (10,000 foot-candles) are co-planar with the normal to the legend. ϕ_1 and ϕ_2 indicate the angle from the normal, from which, the photometer and light source are oriented, respectively.

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Ⓐ Night vision imaging system (NVIS) compatibility. Applicable when specified (see table XII) only for switches with high efficiency (type 2) push button and incandescent illumination

Requirements: The illuminated color and luminance of the legend shall be compatible with NVIS goggles as specified in MIL-L-85762. The NVIS green A color shall be within a circle where $u'_1 = 0.088$, $v'_1 = 0.543$ and the radius = 0.037. The NVIS green B color shall be within a circle where $u'_1 = .131$, $v'_1 = .623$ and the radius = .057 when the legend luminance is set at 0.1 foot-lambert. The NVIS yellow color shall be within a circle where $u'_1 = .274$, $v'_1 = .622$ and the radius = .083 when the legend luminance is set at 15.0 foot-lamberts. Formula IV is solved for the radius (r) to confirm compliance.

$$(u' - u'_1)^2 + (v' - v'_1)^2 \leq r^2 \qquad \text{formula IV}$$

Where.

u', v' = 1976 UCS chromaticity coordinates of the test article.

u'_1, v'_1 = 1976 UCS chromaticity coordinates of the center point of the specified color area above.

r = radius of the allowable circular area on the 1976 UCS chromaticity diagram for the specified color.

The spectral radiance of the legend shall be compatible with NVIS goggles as specified in MIL-L-85762 where NVIS radiance (NR) is defined. The NVIS green A and green B spectral radiance output shall not exceed 1.7×10^{-10} NR for type I, class A and type II, class B goggles when the legend luminance is set at 0.1 foot-lambert. The NVIS yellow spectral radiance output shall not exceed 1.4×10^{-7} NR nor be less than 4.7×10^{-8} NR for type I, class B goggles, and shall not exceed 1.4×10^{-7} NR for type II, class B goggles when the legend luminance is set at 15.0 foot-lamberts. See tables VI and VII.

Test methods: The test procedure for measuring the NVIS luminance, color and spectral radiance shall be in accordance with MIL-L-85762 for illuminated controls (for NVIS green A and B) and warning signals or master caution signals (for NVIS yellow).

The night vision goggle compatible feature, when specified for type S displays, is in addition to the sunlight readable feature and all minimum sunlight readable requirements shall be maintained on all type S displays along with the unique requirements for night vision goggle compatibility

Ⓐ TABLE VI. Illuminated chromaticity limits and luminance for NVIS compatible, sunlight readable displays (type S) - incandescent illumination.

Color	Color code	u' 1/	v' 1/	r 1/	Minimum average character luminance 2/
NVIS green A	H	.088	.543	.037	100 foot-lamberts
NVIS green B	J	131	.623	.057	200 foot-lamberts
NVIS yellow	K	.274	.622	.083	200 foot-lamberts

1/ Chromaticity is expressed as u' and v' coordinates of the 1976 UCS diagram. The values shown describe a circle whose center is at u', v', and of radius (r). The area enclosed by this circle intersecting the spectral locus is the envelope limiting the acceptable color space.

2/ When illuminated by 4 T-1 midget flange base subminiature lamps of .15 ±.01 MSCP at 5 V dc, 2350°K.

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Ⓐ TABLE VII. Illuminated chromaticity limits and luminance for NVIS compatible, type N and type B displays with incandescent illumination.

Color	Color code	u' 1/	v' 1/	r 1/	Character luminance 2/
NVIS green A	H	.088	.543	.037	0.5 to 3.0 foot-lambert
NVIS green B	J	.131	.623	.057	0.5 to 3.0 foot-lamberts

1/ Chromaticity is expressed as u' and v' coordinates of the 1976 UCS diagram. The values shown describe a circle whose center is at u', v', and of radius (r). The area enclosed by this circle intersecting the spectral locus is the envelope limiting the acceptable color space.

2/ When illuminated by 4 T-1 midget flange base subminiature lamps of .15 ±.01 MSCP at 5 V dc, 2350°K.

Touch temperature: Applicable to incandescent illuminated switches. When switches are tested as specified below, the maximum difference between the legend face temperature and the ambient temperature shall be as specified in table VIII.

Ⓐ A single switch shall be mounted in a 2 inch by 3 inch by one eighth of a inch thick aluminum plate, using the recommended panel cutout (see figure 4). All four lamps shall be continuously energized at their maximum rated voltage. The test shall be performed with the switch in still air which has a temperature of 70°F ±5°F. The legend face temperature at the center of the legend area shall be recorded at not less than 5-minute intervals, until the temperature has stabilized. The temperature shall be considered stable when three successive readings taken at 5-minute intervals indicate no change in temperature. The ambient temperature shall be measured during the same period at a point sufficiently far from any heat source (including the test switch) to obtain a representative temperature value for the air reaching the switch by convection. The difference between the legend face temperature and the ambient temperature shall then be calculated by subtracting the lowest recorded ambient temperature from the highest recorded legend face temperature. The test shall be performed with each of the lamp types specified in table VIII.

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Ⓐ TABLE VIII. Touch temperature difference.

Bulbs four each	Short push button standard temperature (type 1)	Short push button low touch option (type 1)	High efficiency push button (type 2)
5 volts 1/ (low power)	50°F	32°F	40°F
5 volts 1/ (high power)	70°F	50°F	40°F
28 volts 1/ (standard)	75°F	52°F	45°F

1/ Lamp definition is:

Description	Voltage	Current	Part number	Life (hours)
Low power	5 V	.06 ampere	685AS15	100,000
High power	5 V	.115 ampere	718AS15	40,000
Standard	28 V	.024 ampere	6839	16,000

QUALITY ASSURANCE

Qualification inspection: All part numbers will be approved for inclusion in the qualified products list upon successful completion of testing specified in table IX. Table IX is based on the use of qualified MIL-S-8805/4 or MIL-S-8805/109 basic switches in the assembly

Ⓐ Group A inspection: See table X.

Ⓐ TABLE IX. Qualification inspection - group submission.

Inspection 1/	Sample 2/
Group I	All samples
Group II	M22885/109WAB06 (2 samples) M22885/109S3234 (2 samples)
Group III	M22885/109WAB06 (2 samples)
Group VI 3/	M22885/109W0106 (2 samples)
Group VII	M22885/109S3234 (2 samples)

See footnotes at end of table.

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Ⓐ TABLE IX. Qualification inspection - group submission - Continued.

Inspection 1/	Sample 2/
Group VIII 4/ 5/	Display type S, high efficiency push button, incandescent lamps. two push buttons of each color (16 samples including NVIS colors) 6/
	Display type W, short push button, standard touch temperature, incandescent lamps: two push buttons of each color (10 samples) 6/
	Display type S 7/, high efficiency push button, LED illumination. two push buttons of each color (6 samples). 6/ 8/

1/ Inspection groups in accordance with qualification inspection table of MIL-S-22885.

2/ Test sample color and legend optional when not specified. For type S displays, legend shall conform to the test requirements for sunlight readability (for high efficiency push buttons only).

3/ Electrical endurance: Test sea level resistive dc only. Lamps used during electrical endurance test shall be industry number 6839 (28 V, .024 A).

4/ Following the field of view test, red color samples of group VIII shall be tested for touch temperature as follows

Short push button, standard touch temperature: 28-volt lamps shall be used.
Short push button, low touch temperature: 5-volt high power lamps shall be used
High efficiency push button 5-volt low power lamps shall be used.

5/ NVIS color and NVIS radiance testing shall be done where applicable during group VIII testing.

6/ Push buttons shall be assembled to any representative switch body during testing

7/ Sunlight readability not applicable when using LED illumination in type S displays.

8/ See application information (table XIV) for LED lighting characteristics

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Ⓐ TABLE X. Group A inspection.

Inspection
Seal
Visual and mechanical examination
Operating characteristics 1/
Dielectric withstanding voltage
Contact resistance 2/
Sunlight readability 3/
NVIS compatibility 4/

1/ Do not exceed low level electrical ratings on gold contact switches.

2/ Contact resistance measurements for gold contact switches shall be measured in accordance with the switch contact resistance of MIL-S-22885, except the test current shall not exceed 10 milliamperes and the open-circuit test voltage shall not exceed 30 millivolts dc. Maximum contact resistance is 3 ohms.

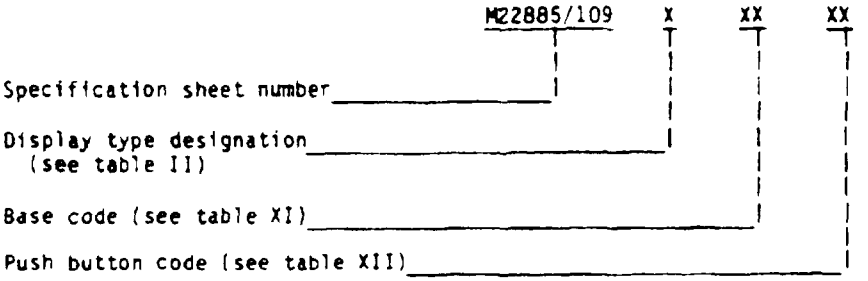
3/ Test one switch per color and use these as visual standards. Visually inspect all switches in each lot against the appropriately colored standard switch. The switches used as visual standards shall be certified annually by performing the sunlight readability test.

4/ Test one switch each for NVIS green A, NVIS green B, and NVIS yellow in accordance with the requirements of MIL-L-85762. These units will serve as standards when viewed with an image intensifier. Visually inspect all NVIS compatible switches in each lot with an image intensifier against the appropriate standard switch. The switches used as visual standards shall be certified annually by performing the NVIS compatibility test.

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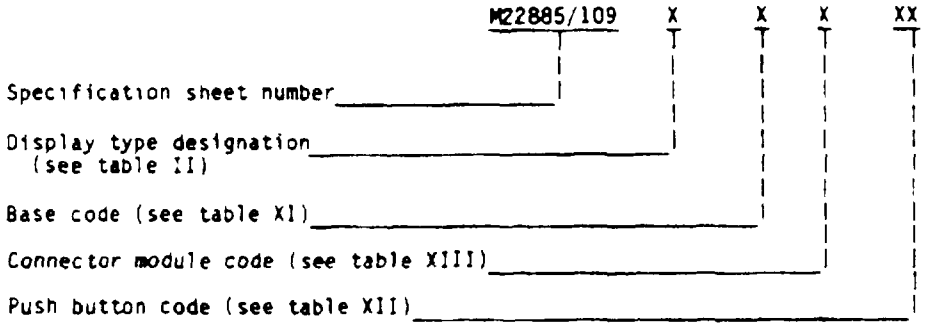
Part numbers:

Assign part numbers for switches with solder lug or PCB pin terminals as follows:



Example M22885/109S1435

Ⓐ Assign part numbers for switches with connector module termination as follows.



Example M22885/109WJF05

Ⓐ Part numbers generated do not include lamps, legends or display color. Switches with legends must be described on a separate altered item drawing in accordance with category II part requirements of MIL-S-22885, including the required color for each segment of the lens. Lamps shall be acquired separately or specified on contract or purchase order.

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Ⓐ TABLE XI. Switch base codes.

Base code	4-pole type 1 1/	4-pole type 2 2/	2-pole type 3 1/	2-pole type 4 2/	Indicator light type 3 1/	Indicator light type 4 2/	Solder terminal	PCB terminal	7-ampere switch	1-ampere switch	Alternate action	Momentary action
For switches with solder or PCB pin terminals												
01	X						X		X		X	
02	X						X		X			X
03	X						X			X	X	
04	X						X			X		X
05	X							X	X		X	
06	X							X	X			X
07	X							X		X	X	
08	X							X		X		X
09		X					X		X		X	
10		X					X		X			X
11		X					X			X	X	
12		X					X			X		X
13		X						X	X		X	
14		X						X	X			X
15		X						X		X	X	
16		X						X		X		X
17			X				X		X		X	
18			X				X		X			X
19			X				X			X	X	
20			X				X			X		X
21			X					X	X		X	
22			X					X	X			X
23			X					X		X	X	
24			X					X		X		X
25				X			X		X		X	
26				X			X		X			X
27				X			X			X	X	
28				X			X			X		X
29				X				X	X		X	
30				X				X	X			X
31				X				X		X	X	
32				X				X		X		X
33					X		X		N/A	N/A	N/A	N/A
34					X			X	N/A	N/A	N/A	N/A
35						X	X		N/A	N/A	N/A	N/A
36						X		X	N/A	N/A	N/A	N/A

See footnotes at end of table.

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Ⓐ TABLE XI. Switch base codes - Continued.

Base code	4-pole type 1 1/	4-pole type 2 2/	2-pole type 3 1/	2-pole type 4 2/	Indicator light type 3 1/	Indicator light type 4 2/	Solder terminal	PCB terminal	7-ampere switch	1-ampere switch	Alternate action	Momentary action
For switches with connector module												
A	X						N/A	N/A	X		X	
B	X						N/A	N/A	X			X
C	X						N/A	N/A		X	X	
D	X						N/A	N/A		X		X
E		X					N/A	N/A	X		X	
F		X					N/A	N/A	X			X
G		X					N/A	N/A		X	X	
H		X					N/A	N/A		X		X
J			X				N/A	N/A	X		X	
K			X				N/A	N/A	X			X
L			X				N/A	N/A		X	X	
M			X				N/A	N/A		X		X
N				X			N/A	N/A	X		X	
P				X			N/A	N/A	X			X
Q				X			N/A	N/A		X	X	
R				X			N/A	N/A		X		X
T					X		N/A	N/A	N/A	N/A	N/A	N/A
V						X	N/A	N/A	N/A	N/A	N/A	N/A

1/ For use with short push button only.

2/ For use with high efficiency push button only.

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Ⓐ TABLE XII. Push button codes.

PB codes 1/	Short PB type 1 2/	Hi efficiency PB type 2 3/	Split common	Single common	NVIS compatible 4/	Standard temperature	Low temperature	Full display	2-way display	3-way display	4-way display
01	X		X			X		X			
02	X		X			X			X		
03	X		X			X				X	
04	X		X			X					X
05	X		X				X	X			
06	X		X				X		X		
07	X		X				X			X	
08	X		X				X				X
17	X			X		X		X			
18	X			X		X			X		
19	X			X		X				X	
20	X			X		X					X
21	X			X			X	X			
22	X			X			X		X		
23	X			X			X			X	
24	X			X			X				X
33		X	X			X	N/A	X			
34		X	X			X	N/A		X		
35		X	X			X	N/A			X	
36		X	X			X	N/A				X
41		X		X		X	N/A	X			
42		X		X		X	N/A		X		
43		X		X		X	N/A			X	
44		X		X		X	N/A				X
49		X	X		X	X	N/A	X			
50		X	X		X	X	N/A		X		
51		X	X		X	X	N/A			X	
52		X	X		X	X	N/A				X
53		X		X	X	X	N/A	X			
54		X		X	X	X	N/A		X		
55		X		X	X	X	N/A			X	
56		X		X	X	X	N/A				X

1/ Codes 09 through 16, 25 through 32, 37 through 40, and 45 through 48 have been canceled and replaced by 01 through 08, 17 through 24, 33 through 36, and 41 through 44 respectively.

2/ Short push button may be used only with base types 1 and 3.

3/ High efficiency push button may be used only with base types 2 and 4.

4/ NVIS compatible displays are available only in display types S, B, and N.

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Ⓐ TABLE XIII. Connector module codes.

Code	Standard mounting sleeve (type HS1)	Mounting sleeve with brackets (type HS2)	Panel thickness		Mounting screws provided
			Min	Max	
A	X		.055	.085	N/A
B	X		.125	.155	N/A
C	X		.205	.235	N/A
D 1/	X		.243	.273	N/A
E		X	.055	.085	MS51959-15B
F		X	.125	.155	MS51959-16B
G		X	.205	.235	MS51959-18B
H 1/		X	.243	.273	MS51959-18B

1/ Not available for short push button (type 1) version.

Ⓐ Application information.

LED lighting characteristics using 4 T-1 submidget flanged base LEDs in high efficiency push buttons with white color filters See table XIV.

Ⓐ TABLE XIV. LED lighting characteristics

Display type	LED color/ wavelength (nm)	Forward voltage V dc	Axial intensity at typical forward voltage MCD	Legend brightness foot lamberts min
S 1/	Red/635 2/	5.0	6.0 - 8.5	5
B	Red/635 2/	5.0	6.0 - 8.5	5
W	Red/635 2/	5.0	6.0 - 8.5	5
C	Red/635 2/	5.0	6.0 - 8.5	5
N	Red/635 2/	5.0	6.0 - 8.5	5
S 1/	Amber/585 3/	5 0	6.0 - 8.5	5
B	Amber/585 3/	5 0	6.0 - 8 5	5
W	Amber/585 3/	5.0	6.0 - 8.5	5
C	Amber/585 3/	5.0	6.0 - 8.5	5
N	Amber/585 3/	5.0	6.0 - 8.5	5
S 1/	Green/565 4/	5.0	6.5 - 9.0	5
B	Green/565 4/	5.0	6.5 - 9.0	5
W	Green/565 4/	5.0	6 5 - 9.0	5
C	Green/565 4/	5.0	6.5 - 9.0	5
N	Green/565 4/	5.0	6.5 - 9.0	5

1/ When using LEDs in type S displays, the resulting display is not sunlight readable.
2/ Red LED: WAMCO (CAGE 58774) part number OLR-108SMF/5V or equivalent.
3/ Amber LED WAMCO (CAGE 58774) part number OLY-108SMF/5V or equivalent
4/ Green LED. WAMCO (CAGE 58774) part number OLG-108SMF/5V or equivalent

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Ⓐ CONCLUDING MATERIAL

Review activities.
Air Force - 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5930-F671)

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